

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

A geologist estimates that the volume of a slab of granite is greater than **12.7** cubic feet but less than **15.7** cubic feet. The geologist also estimates that the slab of granite weighs **165** pounds per cubic foot of volume. Which inequality represents this situation, where x represents the estimated total weight, in pounds, of the slab of granite?

Answer

- A. $165 - 15.7 < x < 165 - 12.7$
- B. $165 + 12.7 < x < 165 + 15.7$
- C. $165(12.7) < x < 165(15.7)$
- D. $\frac{165}{15.7} < x < \frac{165}{12.7}$

Correct Answer: C

Rationale

Choice C is correct. It's given that the estimated volume of the slab of granite is greater than **12.7** cubic feet but less than **15.7** cubic feet. It's also given that the estimated weight per cubic foot of volume of that slab is **165** pounds. The estimated total weight of the slab of granite, in pounds, can be calculated by multiplying the estimated volume by the estimated weight per cubic foot. Therefore, if the estimated volume of the slab of granite is **12.7** cubic feet, its estimated total weight is **165(12.7)** pounds, and if the estimated volume of the slab of granite is **15.7** cubic feet, its estimated total weight is **165(15.7)** pounds. Since the estimated volume of the slab of granite is greater than **12.7** cubic feet but less than **15.7** cubic feet, the estimated total weight x , in pounds, must be greater than **165(12.7)** pounds and less than **165(15.7)** pounds. This situation can be represented by the inequality $165(12.7) < x < 165(15.7)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.